AI/ML Engineer

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Summary:

Experienced AI & Machine Learning Engineer with 5+ years of experience, including **3+ years specializing in Generative AI and LLM-powered solutions** for chatbots, summarization, and document intelligence. Skilled in building **agentic AI systems** with multi-agent workflows, LangChain, LangGraph, OpenAI SDK, MCP, and A2A, as well as **RAG pipelines** leveraging semantic search and vector databases (FAISS, Pinecone, Weaviate, Azure AI Search). Proficient in **fine-tuning LLMs** (GPT-4o, LLaMA-2, Falcon, Mistral) using **LoRA, PEFT**, and advanced **prompt engineering techniques**. Delivered scalable **ML pipelines and AI microservices** for risk scoring, churn prediction, fraud detection, OCR, and sentiment analysis using **Python, PySpark, XGBoost, TensorFlow, and PyTorch**, deployed across **Azure ML, AWS SageMaker, and GCP Vertex AI**. Experienced in **MLOps, CI/CD, and monitoring** with **Azure DevOps, MLflow, Prometheus, Grafana, and Great Expectations**, ensuring robust, production-grade AI systems. Strong foundations in **data science, NLP, computer vision** and cloud-native engineering practices.

Education:

University of Texas at Arlington, TX USA | master's in computer science **January 2023 – December 2024**

DNR College of Engineering and Technology, Bhimavaram, India |Bachelor of Technology in Computer Science **June 2017 – June 2021**

Technical Skills:

Languages	Python, SQL, Bash, JavaScript (basic)
ML & AI Frameworks	scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch, Hugging Face Transformers
Generative AI & LLMs	OpenAI API (GPT-4o, GPT-3.5) , LLaMA-2, Falcon, Mistral, Stable Diffusion, LangChain, LlamaIndex, LangGraph, CrewAI, OpenAI Agents SDK, RAG pipelines
Model Tuning & Training	LoRA, PEFT, Hugging Face, Prompt Engineering, Transfer Learning (few-shot, CoT, zero-shot, structured)
Data Processing & NLP	Pandas, NumPy, spaCy, NLTK, OpenCV, BERT, RoBERTa, Text Classification, Named Entity Recognition, LLM fine-tuning, Prompt Engineer
Deployment & serving	FastAPI, Flask, Docker, Kubernetes, ONNX, Torch Script
Cloud Platforms	Azure Machine Learning, Azure Blob Storage, AWS(Lambda, EC2, S3) Google Cloud Storage, Vertex AI
Experiment Tracking	MLflow, Weights & Biases, Tensor Board
Vector Search & Databases	FAISS, Chroma, Pinecone, Weaviate, Azure AI Search, PostgreSQL, MongoDB, SQLite, Redis
DevOps & MLOps	Git, GitHub Actions, Kubernetes, Docker, CI/CD, Terraform (basic) ,JIRA, Confluence, Agile/Scrum
Visualization	matplotlib, seaborn, Plotly, Power BI (basic)
Other Tools	Jupyter, VS Code, Azure Notebooks, Google Collab

Client: T- Mobile
Role: GEN AI Engineer **February 2025 - Present**

Responsibilities:

- Designed and deployed **Agentic AI systems** using **LangChain**, **LangGraph**, and **OpenAI Agents SDK**, enabling tool-using LLMs with **dynamic function-calling** and **stateful memory** for autonomous task execution.
- Developed and **launched domain-specific chatbots** powered by **GPT-4o** and **LLaMA-2**, integrating with **FastAPI** and structured **prompt templates** for scalable, consistent user interaction.
- Built **Retrieval-Augmented Generation (RAG) pipelines** with **Azure AI Search**, **FAISS**, and **Pinecone**, enhancing contextual retrieval and **improving response accuracy by 35%**.
- **Fine-tuned LLMs** (LLaMA, Falcon, Mistral) using **LoRA**, **PEFT**, and **Hugging Face Transformers**, optimizing latency and performance for **enterprise workloads**.
- Implemented **advanced prompt engineering** strategies—few-shot, zero-shot, and chain-of-thought—to improve reasoning depth and reduce hallucinations across production models.

- Developed inference microservices and **LLM APIs** using **FastAPI** and **Spring Boot WebFlux**, enabling real-time model interaction across multiple backend systems.
- Automated **full MLOps pipelines** with **Git**, **Azure DevOps**, and **MLflow**, supporting continuous training, deployment, and monitoring of GenAI models in production.
- **Led GenAI adoption initiatives** across **Azure and AWS**, containerizing models with **Docker** and **Kubernetes**, and mentoring teams on **LLM integration and RAG architecture**.

Environment: Python, SQL, GPT-4o/3.5, LLaMA-2, Falcon, Mistral, Hugging Face Transformers, PyTorch, LangChain, LangGraph, RAG pipelines, Prompt Engineering, FAISS, Pinecone, Weaviate, Azure AI Search, FastAPI, Flask, Spring Boot WebFlux, Docker, Kubernetes (AKS/EKS), Azure ML, Azure OpenAI, AWS SageMaker, GCP Vertex AI, Azure DevOps, Git, CI/CD, MLflow, Prometheus, Grafana.

Client: Cardinal Health
Role: AI /ML Engineer **January 2023 – December 2024**

Responsibilities:

- **Engineered and automated** end-to-end AI/ML pipelines for risk scoring, churn prediction, fraud detection, telematics underwriting, and sentiment analysis using Python, PySpark, scikit-learn, XGBoost, and Databricks.
- Built **predictive and generative AI solutions** with **PyTorch, TensorFlow, Keras, and Hugging Face**, implementing **LLM-based features** (GPT-4o, LLaMA-2, Falcon, Mistral) using **LangChain, LangGraph, and AWS Bedrock**, with fine-tuning via **LoRA and PEFT**.
- Developed **NLP and voicebot pipelines** for NER, summarization, translation, and multilingual chatbots using **spaCy, BERT/RoBERTa, and Dialogflow**, boosting efficiency by 40%.
- Architected **RAG pipelines and semantic search** with **FAISS, Pinecone, Weaviate, and OpenSearch**, enabling context-aware retrieval and tool-based reasoning.
- Delivered **cloud-native microservices and APIs** with **FastAPI, Flask, and Spring Boot**, containerized via **Docker**, orchestrated on **Kubernetes (AKS/EKS)**, and deployed on **AWS SageMaker, Azure ML, ECS, and Lambda**.
- Implemented **MLOps workflows** with **Azure DevOps, GitHub Actions, MLflow, Great Expectations, Prometheus, and Grafana**, ensuring continuous training, monitoring, and governance of production AI systems.

Environment: Python, SQL, PySpark, scikit-learn, XGBoost, TensorFlow, Keras, PyTorch, Hugging Face Transformers, spaCy, BERT/RoBERTa, GPT-4o, LLaMA-2, Falcon, Mistral, LangChain, LangGraph, AWS Bedrock, RAG, FAISS, Pinecone, Weaviate, OpenSearch, FastAPI, Flask, Spring Boot, React.js, Next.js, Streamlit, Docker, Kubernetes, AWS SageMaker, ECS, Lambda, Azure ML, Azure Functions, Azure Cognitive Services, GCP Vertex AI, Azure DevOps, GitHub Actions, Terraform, CloudFormation, MLflow, Prometheus, Grafana, Git.

Client: LTI Mindtree
Role: ML Engineer **July 2020 – December 2022**

Responsibilities:

- Built and optimized **deep learning solutions** for image classification, OCR, and sequence modeling across insurance and finance domains using TensorFlow, Keras, and PyTorch.
- Developed advanced **NLP pipelines** for NER, sentiment analysis, and text classification with spaCy, NLTK, Hugging Face Transformers, and embeddings, alongside a YOLO-based CV pipeline that achieved 95% accuracy in exam proctoring anomaly detection.
- Engineered **end-to-end Azure ML projects**, leveraging Azure ML, Azure Functions, and Cognitive Services with Azure DevOps for scalable deployments, monitoring, and CI/CD automation.
- Created and deployed **AI microservices** with Flask, FastAPI, and Spring Boot WebFlux, integrated with Kafka for real-time event handling, and applied interpretability tools (SHAP, LIME, Fairlearn) for responsible AI, supported by reusable Jupyter workflows for EDA, feature engineering, and visualization.
- Designed reusable **Python/Jupyter workflows** for EDA, feature engineering, hyperparameter tuning, and visualization with **Pandas, NumPy, Seaborn, and Matplotlib**.

Environment: Pandas, NumPy, Seaborn, Matplotlib, Jupyter, scikit-learn, Python, SQL, TensorFlow, Keras, PyTorch, Hugging Face Transformers, spaCy, NLTK, YOLO, Azure ML, Azure Functions, Azure Cognitive Services, Azure DevOps, Flask, FastAPI, Spring Boot WebFlux, Kafka, Git

Projects:

Enterprise Knowledge Assistant (RAG + Agentic AI)

- Built an enterprise-scale document intelligence system using **LangChain**, **LangGraph**, and **GPT-4o**, integrating **multi-agent orchestration** and **FAISS/Azure AI Search-based RAG pipelines** to achieve **92% factual accuracy** and reduce data retrieval time by **60%**.